1 What is SAP?

SAP stands for **Systems, Applications and Products in Data Processing**.

SAP is an **ERP (Enterprise Resource Planning) software** used by companies to manage all their business processes in one integrated system.

SAP software was developed by **SAP SE**.

Before SAP, companies used **different software for different departments**, which created many problems. SAP solved this by integrating everything into one system.

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2 What is ERP?

ERP means Enterprise Resource Planning.

ERP is a system used to manage all business departments together.

Departments in a company:

- Finance
- Sales
- Human Resources
- Inventory
- Production
- Purchasing

ERP integrates all these departments into **one system**.

Example ERP systems:

- **SAP S/4HANA**
- **Oracle ERP**
- **Microsoft Dynamics 365**

SAP is the **most widely used ERP system** in the world.

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3 Problems Before SAP (Real-Time Scenario)

Before SAP, companies used **separate systems**.

Example company departments:

Finance Department

Used accounting software like **Tally**.

HR Department

Employee details stored in **Microsoft Excel**.

Sales Department

Used billing software or manual systems.

Inventory Department

Used local database or separate software.

Real-Time Problem

Customer orders **100 laptops**.

Process before SAP:

1. Sales team writes order.
2. They inform warehouse manually.
3. Warehouse checks stock manually.
4. Finance creates invoice separately.
5. Management checks reports later.

Problems:

- Data mismatch
- Manual communication
- Time delay
- Errors in data

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4 How SAP Solved This Problem (Real-Time Example)

Customer orders **100 laptops**.

In SAP system:

1 Sales team creates **Sales Order**
2 SAP automatically checks **Inventory stock**
3 SAP creates **Delivery document**
4 Finance automatically gets **Invoice data**
5 Management sees **real-time reports**

Everything happens in **one system automatically**.

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5 Why Companies Use SAP

Companies use SAP because it helps to:

1. Integrate all departments
2. Reduce manual work
3. Provide real-time data
4. Improve business efficiency
5. Reduce errors
6. Improve reporting and analytics

Example companies using SAP:

- Tata Group
- Reliance Industries
- Infosys

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6 SAP Modules

SAP has different modules for different departments.

SAP FICO

Finance and accounting.

SAP MM

Material management and inventory.

SAP SD

Sales and distribution.

SAP HCM

Human resource management.

SAP ABAP

Programming language used to develop SAP applications.

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7 Real-Time Example (Full Flow)

Customer orders a **mobile phone** from a company.

SAP process:

1. Sales creates **Sales Order (SD Module)**
2. Warehouse checks **Stock (MM Module)**
3. Delivery is created
4. Finance generates **Invoice (FICO Module)**

Payment is recorded

All data flows automatically inside SAP.

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8 SAP Database

SAP uses a powerful database called **SAP HANA**.

It stores and processes data very fast.

⬙ What is SAP Fiori?

SAP Fiori is a **modern user interface (UI)** used to access SAP applications easily.

Developed by **SAP SE**

Simple:
Fiori = Easy + User-friendly SAP screens (apps)

⬙ Why SAP Fiori?

Old SAP (**SAP GUI**) problems:

- Complex screens
- Hard to use
- Not mobile friendly

Fiori solution:

- Simple design
- Mobile + Desktop support
- Easy navigation

🔹 How Fiori Works?

1. User logs into Fiori Launchpad
2. Apps (tiles) are shown based on role
3. User clicks app
4. App connects to backend (**SAP S/4HANA**)
5. Data is displayed

🔹 Key Concept

Role-based apps

Example:

- Manager → Reports
- Employee → Apply Leave
- Sales → Create Order

🔹 Real-Time Example

Leave Application

Old way (SAP GUI):

- Enter T-code
- Complex screen

Fiori way:

- Click "Apply Leave" app
- Fill form
- Submit

Easy & fast

🔹 Types of Fiori Apps

1. Transactional → Work (Create, Update)
2. Analytical → Reports, Charts
3. Fact Sheet → Display details

◈ Advantages

- User-friendly
- Faster
- Mobile support
- Less training required

◈ One Line

SAP Fiori = Simple apps to use SAP easily

What is SAP BTP?

SAP BTP (Business Technology Platform) is a **cloud platform** used to build, extend, and integrate SAP applications.

Developed by **SAP SE**

Simple:
BTP = Platform to build apps outside SAP core

◈ Why SAP BTP?

Before BTP:

- Custom code directly in SAP system
- System becomes complex
- Upgrades become difficult

BTP Solution:

- Keep SAP core clean
- Build custom apps in cloud

◈ Key Idea

Clean Core Concept

- Core SAP → Standard
- Custom Development → BTP

What We Do in BTP?

Using BTP, we can:

- Build applications
- Create APIs
- Integrate systems
- Store and manage data
- Do analytics

How BTP Works?

1. User accesses application
2. App runs on BTP (cloud)
3. Connects to backend (**SAP S/4HANA**)
4. Data is fetched/updated

Technologies in BTP

- **CAP (Cloud Application Programming Model)**
- **Node.js / Java**
- OData Services
- SAP Fiori (UI)

Real-Time Example

Scenario: Leave Approval App

Instead of developing inside SAP:

1. Build app in BTP
2. Connect to SAP system
3. Employee applies leave
4. Manager approves

SAP core remains clean

◈ Advantages of BTP

- Keeps system clean
- Cloud-based (no local setup)
- Scalable
- Faster development
- Easy integration

◈ One Line

SAP BTP = Cloud platform to build and extend SAP applications